DEPARTMENT OF MATHEMATICS - COURSE SYLLABUS

Course Code: MATH-301: Advanced Linear Algebra & Spectral Theory

Module 1: Vector Spaces, Subspaces, Basis, Dimension (8 Hours)

Module 2: Linear Transformations, Kernel, Rank-Nullity Theorem (10 Hours)

Module 3: Eigenvalues, Eigenvectors, Cayley-Hamilton Theorem (12 Hours)

Module 4: Inner Product Spaces, Gram-Schmidt Orthogonalization (10 Hours)

Reference Text: Hoffman & Kunze, Linear Algebra 2nd Ed., ISBN 978-0135367971